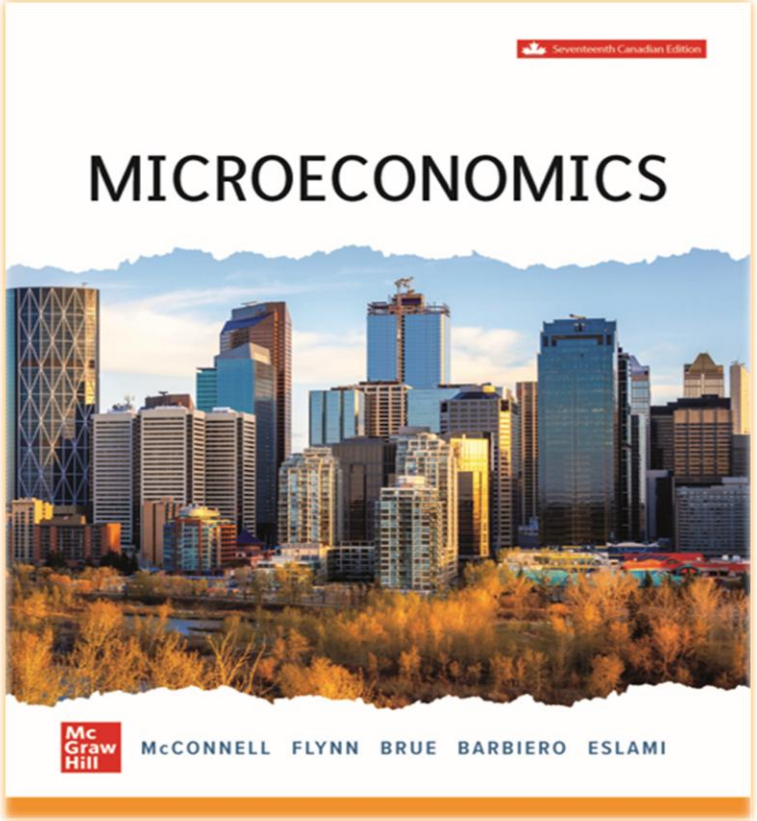
CHAPTER 6

Elasticity



PowerPoint Presentation Prepared by:
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LEARNING OBJECTIVES

LO6.1 Explain and calculate the price elasticity of demand.

LO6.2 Explain the usefulness of the total-revenue test.

LO6.3 List the factors that affect the price elasticity of demand.

LO6.4 Describe and apply the price elasticity of supply.

LO6.5 Apply cross elasticity of demand and income elasticity of demand.

LO6.6 Apply elasticity to tax incidence.

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INTRODUCTION

- Elasticity measures how changes in price and income affect supply and demand.
- It explains why buyers greatly reduce purchases of some goods when prices rise.
- It shows why demand for certain products increases more with higher income.
- Responses to price and income changes can be substantial or minimal.

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6.1 PRICE ELATICITY OF DEMAND 1/12

Law of Demand

- An increase in price causes a decrease in quantity demanded (and vice-versa)
- But how much does quantity demanded change in response to a change in price?
  - Elasticity gives us a measure of the responsiveness (or sensitivity) of quantity demanded by consumers to a change in price

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6.1 PRICE ELATICITY OF DEMAND 2/12

The Price Elasticity Coefficient and Formula

$$E_d = \frac{\text{Percentage change in quantity demanded of product X}}{\text{Percentage change in the price of product X}} = \frac{\% \Delta Q_d}{\% \Delta P}$$

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6.1 PRICE ELATICITY OF DEMAND 3/12

The Price Elasticity Coefficient and Formula

$$E_d = \frac{\text{Percentage change in quantity demanded of product X}}{\text{Percentage change in the price of product X}} = \frac{\% \Delta \boldsymbol{Q_d}}{\% \Delta P}$$

– When Q_d responds strongly to a change in P, demand is elastic

– When Q_d responds weakly to a change in P, demand is inelastic

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6.1 PRICE ELATICITY OF DEMAND 4/12

The Price Elasticity Coefficient and Formula

- If the quantity demanded increased from 4 to 5 units the percentage change would be:

$$\%\Delta Q_d = \Delta Q_d / Q_0 = \tfrac{1}{4} \text{ x } 100 = 25\%$$

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6.1 PRICE ELATICITY OF DEMAND 5/12

The Price Elasticity Coefficient and Formula

– If the quantity demanded increased from 4 to 5 units the percentage change would be:

$$\%\Delta \boldsymbol{Q_d} = \Delta \boldsymbol{Q_d}/\boldsymbol{Q_0} = \tfrac{1}{4} \text{ x } \mathbf{100} = \mathbf{25\%}$$

– If the quantity demanded dropped from 5 to 4, the percentage change would be:

$$\%\Delta \boldsymbol{Q_d} = \Delta \boldsymbol{Q_d}/\boldsymbol{Q_0} = \tfrac{1}{5} \text{ x } 100 = 20\%$$

– Should we use 25 percent or 20 percent in the numerator of the elasticity formula?

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The Price Elasticity Coefficient and Formula: Using Averages

- To avoid confusion about start and end points, we use average change in Q_d.
- For the same change in quantity demanded between 4 to 5 units the average percentage change would be:

$$\%\Delta Q_d = \frac{\Delta Q_d}{\left(\frac{Q_0+Q_1}{2}\right)} = \frac{1}{\left(\frac{Q_0+Q_1}{2}\right)} = \frac{1}{(5)} = 0.222 \; or \; 22.2\%$$

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The Price Elasticity Coefficient and Formula: Using Averages

- Price Elasticity of Demand (Midpoint) Formula:

$$E_d = \frac{\dfrac{\text{Change in quantity}}{\left(\dfrac{\text{sum of quantities}}{2}\right)}}{\dfrac{\text{Change in price}}{\left(\dfrac{\text{sum of prices}}{2}\right)}}$$

Or,

$$E_d = \left[\frac{\text{change in quantity}}{\left(\dfrac{\text{sum of quantities}}{2}\right)}\right] \div \left[\frac{\text{change in price}}{\left(\dfrac{\text{sum of prices}}{2}\right)}\right]$$

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The Price Elasticity Coefficient and Formula: Using Averages

- Price Elasticity of Demand (Midpoint) Formula:

$$E_d = \left[\frac{\text{change in quantity}}{\left(\frac{\text{sum of quantities}}{2}\right)}\right] \div \left[\frac{\text{change in price}}{\left(\frac{\text{sum of prices}}{2}\right)}\right]$$

- **Example:** Suppose quantity demanded increases from 4 to 5 units when the price falls from \$20 to \$10.

$$E_d = \frac{\Delta Q}{\left(\frac{Q_0+Q_1}{2}\right)} \div \frac{\Delta P}{\left(\frac{P_0+P_1}{2}\right)}$$

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6.1 PRICE ELATICITY OF DEMAND 9/12

The Price Elasticity Coefficient and Formula: Using Averages

- Price Elasticity of Demand (Midpoint) Formula:

$$E_d = \left[\frac{\text{change in quantity}}{\left(\frac{\text{sum of quantities}}{2}\right)}\right] \div \left[\frac{\text{change in price}}{\left(\frac{\text{sum of prices}}{2}\right)}\right]$$

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$$E_d = \frac{\Delta Q}{\left(\frac{Q_0+Q_1}{2}\right)} \div \frac{\Delta P}{\left(\frac{P_0+P_1}{2}\right)}$$

$$E_d = \frac{\Delta Q}{\left(\frac{Q_0+Q_1}{2}\right)} \div \frac{\Delta P}{\left(\frac{P_0+P_1}{2}\right)} = \frac{1}{\left(\frac{4+5}{2}\right)} \div \frac{10}{\left(\frac{10+20}{2}\right)}$$

$$E_d = 0.33$$

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6.1 PRICE ELATICITY OF DEMAND 10/12

The Price Elasticity Coefficient and Formula: Using Percentages & Eliminating the Minus Sign

- Why percentages rather than absolute amount?
  - Unit free measure
  - Easier to compare responsiveness across products
- Eliminate the minus sign
  - E_d will always be negative because price and quantity are inversely related
  - Easier to compare elasticities

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6.1 PRICE ELATICITY OF DEMAND 11/12

Interpretation of Elasticity of Demand

- $E_d > 1$ demand is elastic
- $E_d = 1$ demand is unit elastic
- $E_d < 1$ demand is inelastic

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6.1 PRICE ELATICITY OF DEMAND 12/12

Interpretation of Elasticity of Demand: Extreme cases

- Perfectly inelastic
  - When a price change results in no change in the quantity demanded
- Perfectly elastic
  - When a small price reduction causes buyers to increase their purchases from zero to all they can obtain, the elasticity coefficient is infinite (= ∞),

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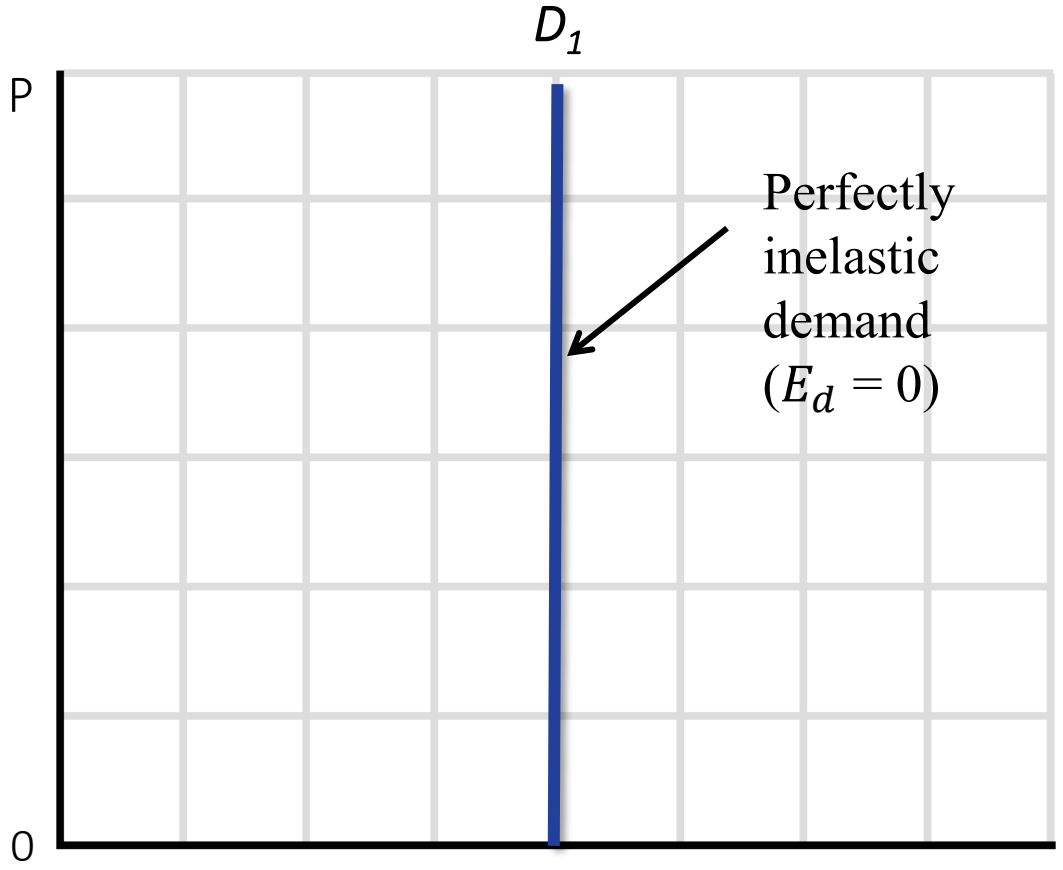
FIGURE 6-1 Perfectly Inelastic and Perfectly Elastic Demand 1/2

- Demand curve D_1 represents perfectly inelastic demand ($E_d = 0$).
  - A price increase does not change the quantity demanded.



(a) Perfectly inelastic demand

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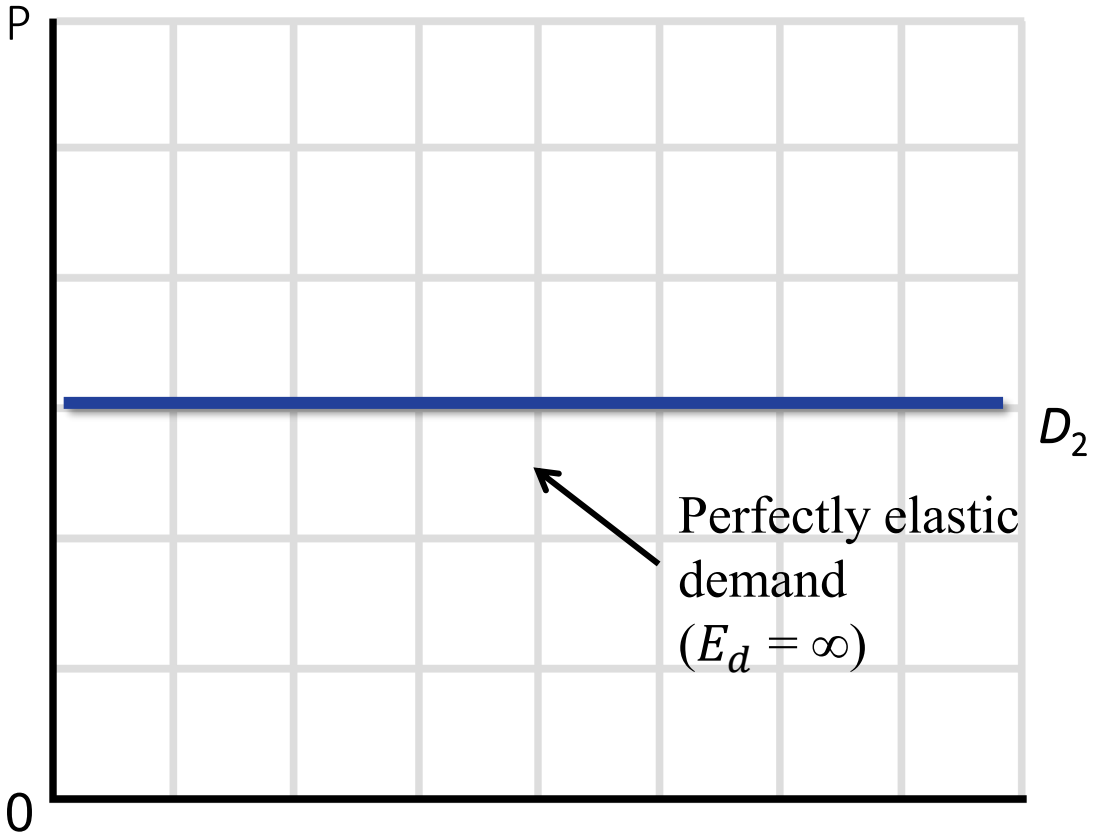
FIGURE 6-1 Perfectly Inelastic and Perfectly Elastic Demand 2/2

- Demand curve D_2 in panel (b) represents perfectly elastic demand.
  - A price increase causes quantity demanded to decline from an infinite amount to zero ($E_d = \infty$).



(b) Perfectly elastic demand

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6.2 THE TOTAL-REVENUE TEST 1/2

Total Revenue (TR)

- TR = P × Q
- TR and E_d are related:
  - If TR changes in the opposite direction from price, demand is elastic
  - If TR changes in the same direction as price, demand is inelastic
  - If TR does not change when price changes, demand is unit elastic

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FIGURE 6-2 The Total-Revenue Test for Price Elasticity 1/3

Demand is elastic

- Price declines from $2 to $1, and total revenue increases from $20 to $40.
- The gain in revenue (blue area) exceeds the loss of revenue (yellow area).

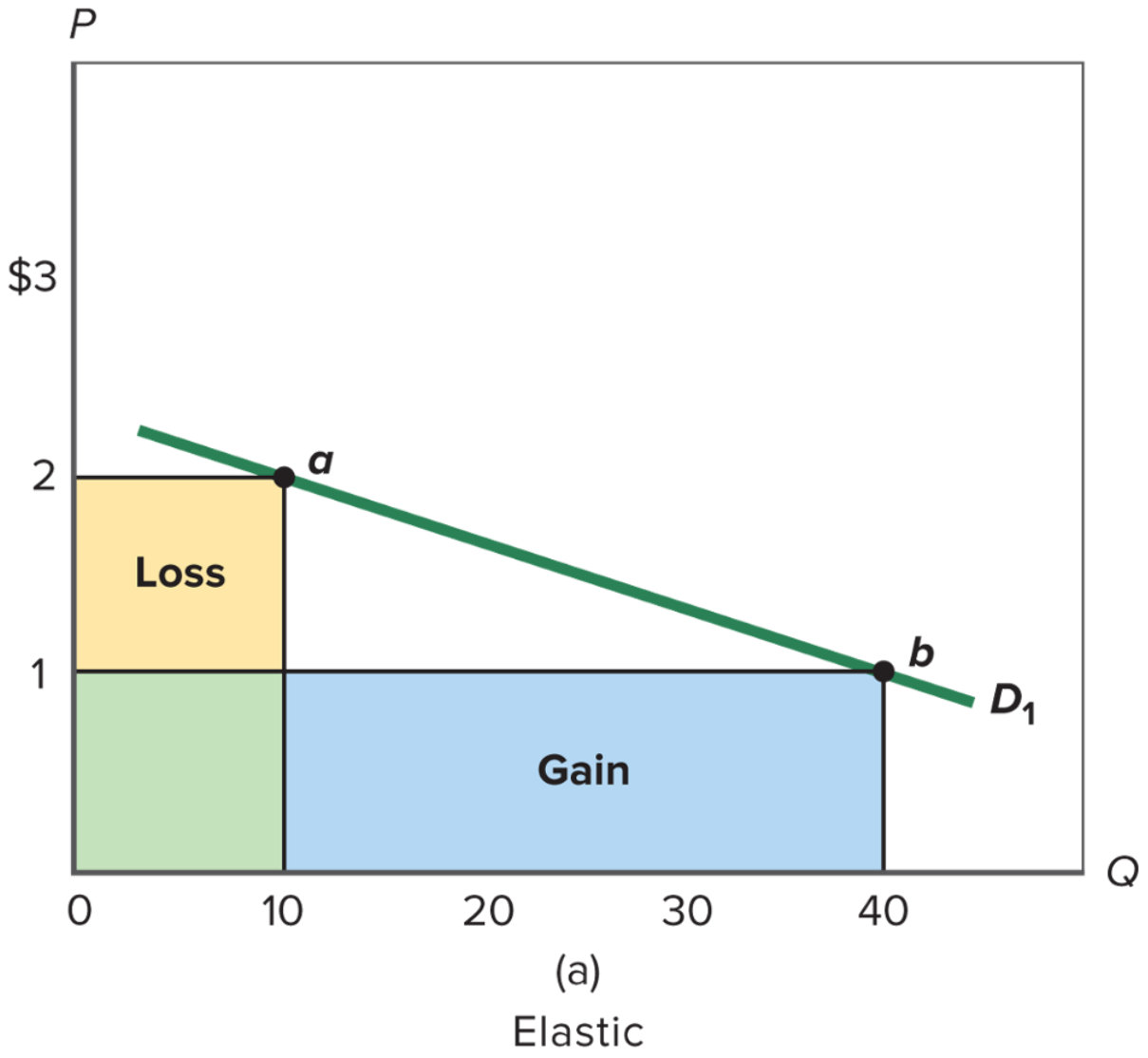

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FIGURE 6-2 The Total-Revenue Test for Price Elasticity 2/3

Demand is inelastic.

- Price declines from $4 to $1, and total revenue falls from $40 to $20.
- The gain in revenue (blue area) is less than the loss of revenue (yellow area).

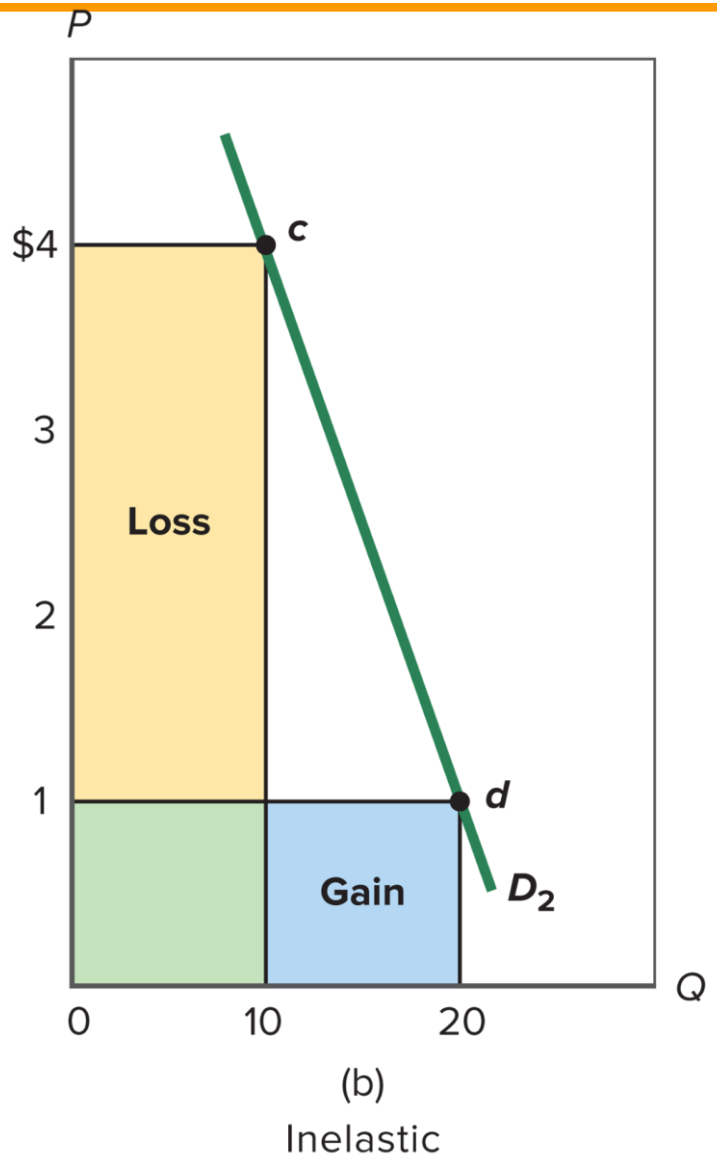

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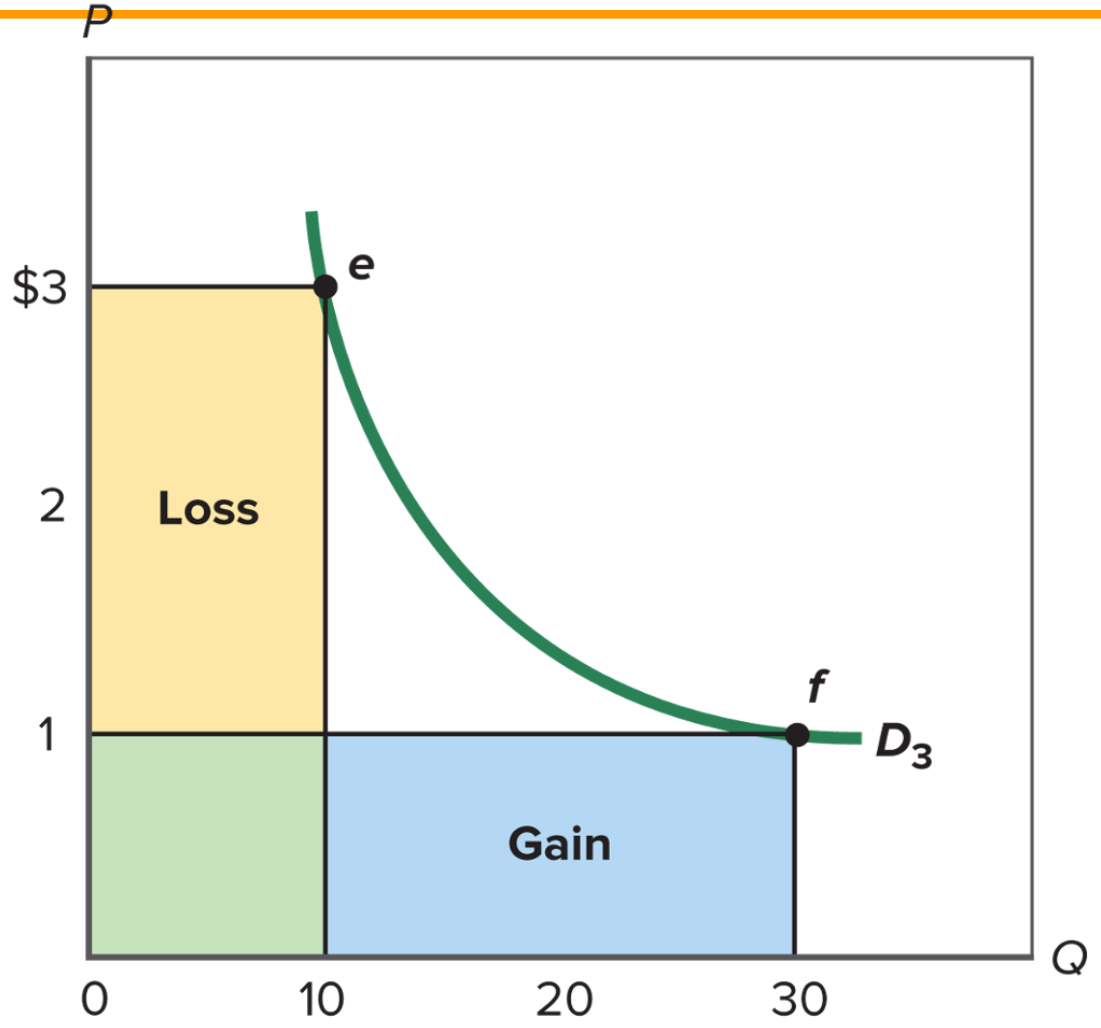
FIGURE 6-2 The Total-Revenue Test for Price Elasticity 3/3

Demand is unit-elastic.

- Price declines from $3 to $1, and total revenue does not change.
- The gain in revenue (blue area) equals the loss of revenue (yellow area).



(c)
Unit-elastic

6.2 THE TOTAL-REVENUE TEST 2/2

Price Elasticity along a Linear Demand Curve

- For all straight-line and most other demand curves
- Demand is more elastic toward the upper left
- Demand is less elastic toward the lower right

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Table 6-1 Price Elasticity of Demand for Movie Tickets as Measured by the Elasticity Coefficient and the Total-Revenue Test

(1) Total Quantity of Tickets Demanded per Week, Thousands	(2) Price per Ticket	(3) Elasticity Coefficient (E_d)	(4) Total Revenue (1) X (2)	(5) Total Revenue Test
1	$8		$8,000	
2	7	5.00	14,000	**Elastic**
3	6	2.60	18,000	**Elastic**
4	5	1.57	20,000	**Elastic**
5	4	1.00	20,000	**Unit Elastic**
6	3	0.64	18,000	**Inelastic**
7	2	0.38	14,000	**Inelastic**
8	1	0.20	8,000	**Inelastic**

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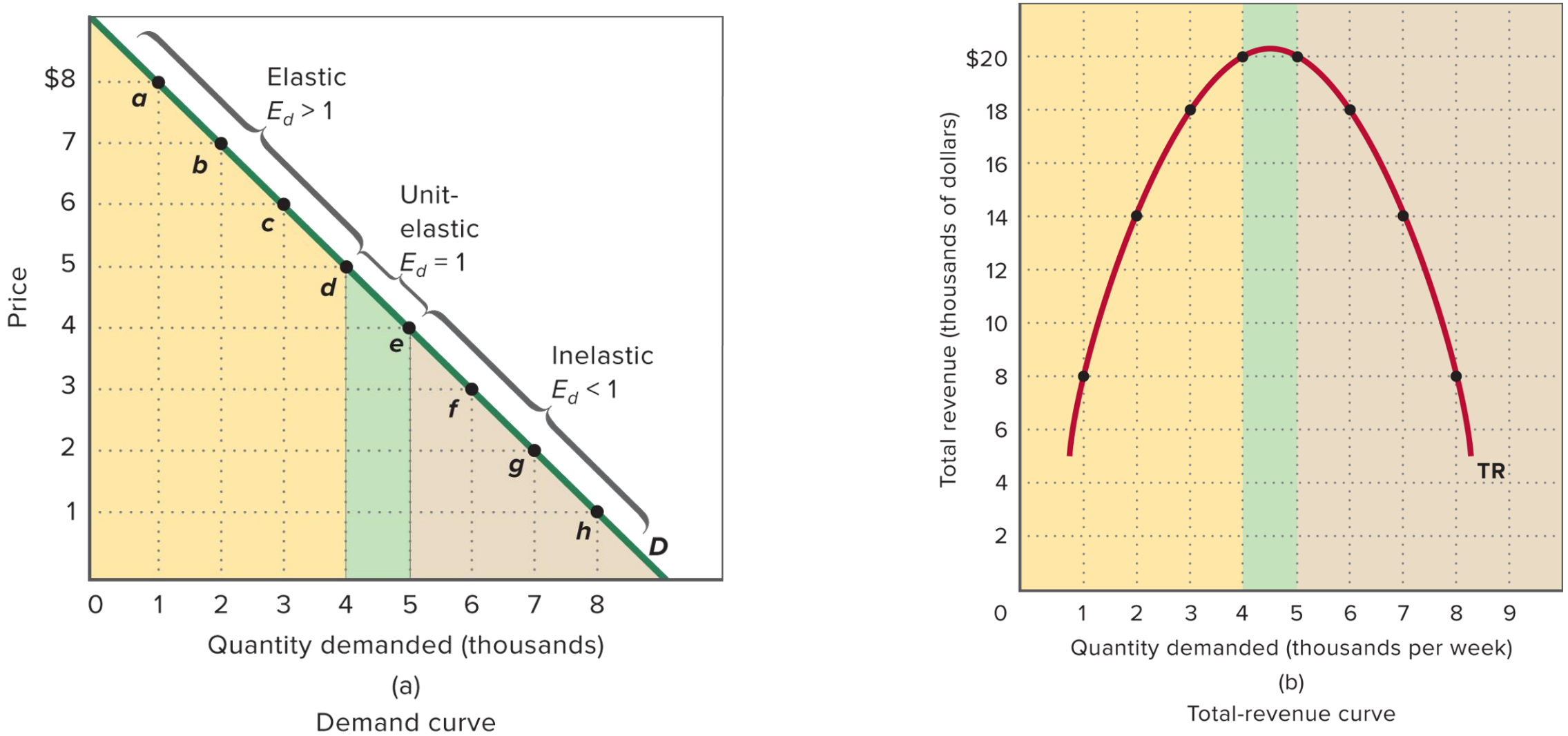
FIGURE 6-3 The Relation Between Price Elasticity of Demand for Movie Tickets and Total Revenue



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7	2	0.38	14,000	**Inelastic**
8	1	0.20	8,000	**Inelastic**

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TABLE 6-2 Price Elasticity of Demand: A Summary

Absolute value of elasticity coefficient	Demand is	Description	Impact on total revenue of a	
			Price increase	Price decrease
Greater than 1 ($E_d > 1$)	Elastic or relatively elastic	Quantity demanded changes by a larger percentage than does price	**Total revenue decreases**	**Total revenue increases**
Equal to 1 ($E_d = 1$)	Unit or unitary elastic	Quantity demanded changes by the same percentage as does price	**Total revenue is unchanged**	**Total revenue is unchanged**
Less than 1 ($E_d < 1$)	Inelastic or relatively inelastic	Quantity demanded changes by a smaller percentage than does price	**Total revenue increases**	**Total revenue decreases**

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6.3 DETERMINANTS OF PRICE ELASTICITY OF DEMAND 1/6

Substitutability

- The larger the number of substitute goods available, the greater the price elasticity of demand.
- Depends on how narrowly the product is defined.
- For example, demand for Reebok is more elastic than for all shoes.

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6.3 DETERMINANTS OF PRICE ELASTICITY OF DEMAND 2/6

Proportion of Income

- The greater the proportion of income spent on a good, the greater the price elasticity of demand for it.
- E.g. chewing gum.

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6.3 DETERMINANTS OF PRICE ELASTICITY OF DEMAND 3/6

Luxuries versus Necessities

- The more a good is considered a luxury rather than a necessity, the greater the price elasticity of demand.
- Easier to do without.

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6.3 DETERMINANTS OF PRICE ELASTICITY OF DEMAND 4/6

Time

- Product demand is more elastic the longer the period under consideration.
- Consumers often need time to adjust to changes in prices.

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Table 6-3 Selected Price Elasticities of Demand

Product or service	Coefficient of price elasticity of demand, E_d	Product or service	Coefficient of price elasticity of demand, E_d
Newspapers	0.10	Household appliances	0.63
Electricity (household)	0.13	Movies	0.87
Bread	0.15	Beer	0.90
Major league baseball tickets	0.23	Shoes	0.91
Telephone service	0.26	Motor vehicles	1.14
Sugar	0.30	Beef	1.27
Eggs	0.32	China, glassware, tableware	1.54
Legal services	0.37	Residential land	1.60
Automobile repair	0.40	Restaurant meals	2.27
Clothing	0.49	Lamb and mutton	2.65
Gasoline	0.60	Fresh peas	2.83
Milk	0.63		

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6.3 DETERMINANTS OF PRICE ELASTICITY OF DEMAND 5/6

Applications of Price Elasticity of Demand

- Large Crop Yields:
  - For farm products with inelastic demand (Ed ≈ 0.20-0.25), increased supply lowers prices and farmers' incomes, suggesting that restricting farm output may be necessary to raise total farm income.
- Sales Taxes:
  - Governments prefer taxing products with inelastic demand (e.g., liquor, gasoline) since higher taxes on elastic products reduce sales and overall tax revenue

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6.3 DETERMINANTS OF PRICE ELASTICITY OF DEMAND 6/6

Applications of Price Elasticity of Demand

- Decriminalization of Drugs:
  - Legalization could lower drug prices, increasing consumption. Proponents argue addicts' inelastic demand means little increase, while opponents claim the elastic demand of casual users would significantly boost overall consumption and social costs

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6.4 PRICE ELASTICITY OF SUPPLY 1/5

Price Elasticity of Supply

- The ratio of the percentage change in quantity supplied of a product to the percentage change in its price.

$$E_s = \frac{\textit{Percentage change in quantity supplied of product X}}{\textit{Percentage change in the price of product X}} = \frac{\%\Delta Q_s}{\%\Delta P}$$

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6.4 PRICE ELASTICITY OF SUPPLY 2/5

Price Elasticity of Supply

- **Elasticity of Supply:**
  - Supply is elastic if the quantity supplied responds significantly to price changes; inelastic if it responds minimally.
- Coefficient (Es):
  - Calculated using the percentage change in quantity supplied over the percentage change in price.

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6.4 PRICE ELASTICITY OF SUPPLY 3/5

Price Elasticity of Supply

- Elasticity Values:
  - Supply is elastic if $Es > 1$, inelastic if $Es < 1$, and unit elastic if $Es = 1$.
- Resource Shift:
  - Supply elasticity depends on how quickly producers can shift resources, increasing with more time to adjust (immediate, short run, long run).

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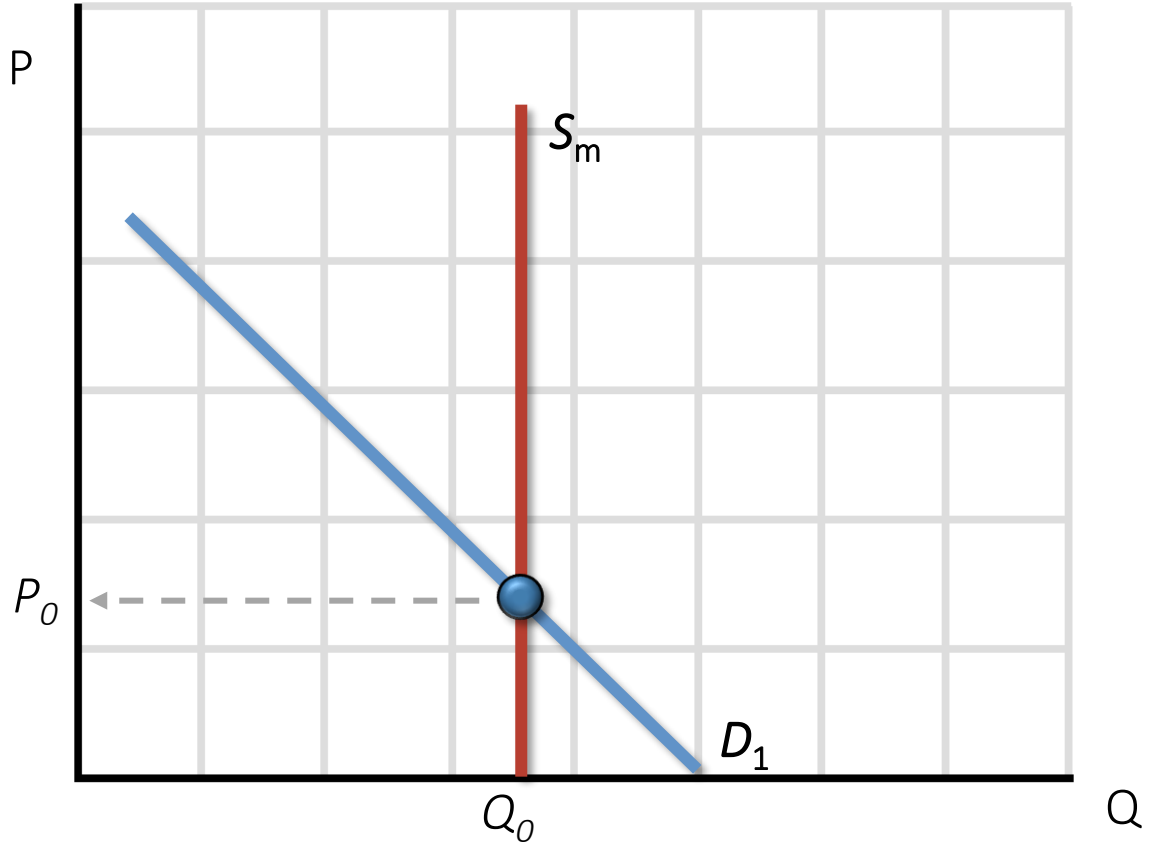
FIGURE 6-4 Time and Elasticity of Supply 1/6



Perfectly inelastic supply

(a) Immediate Market Period

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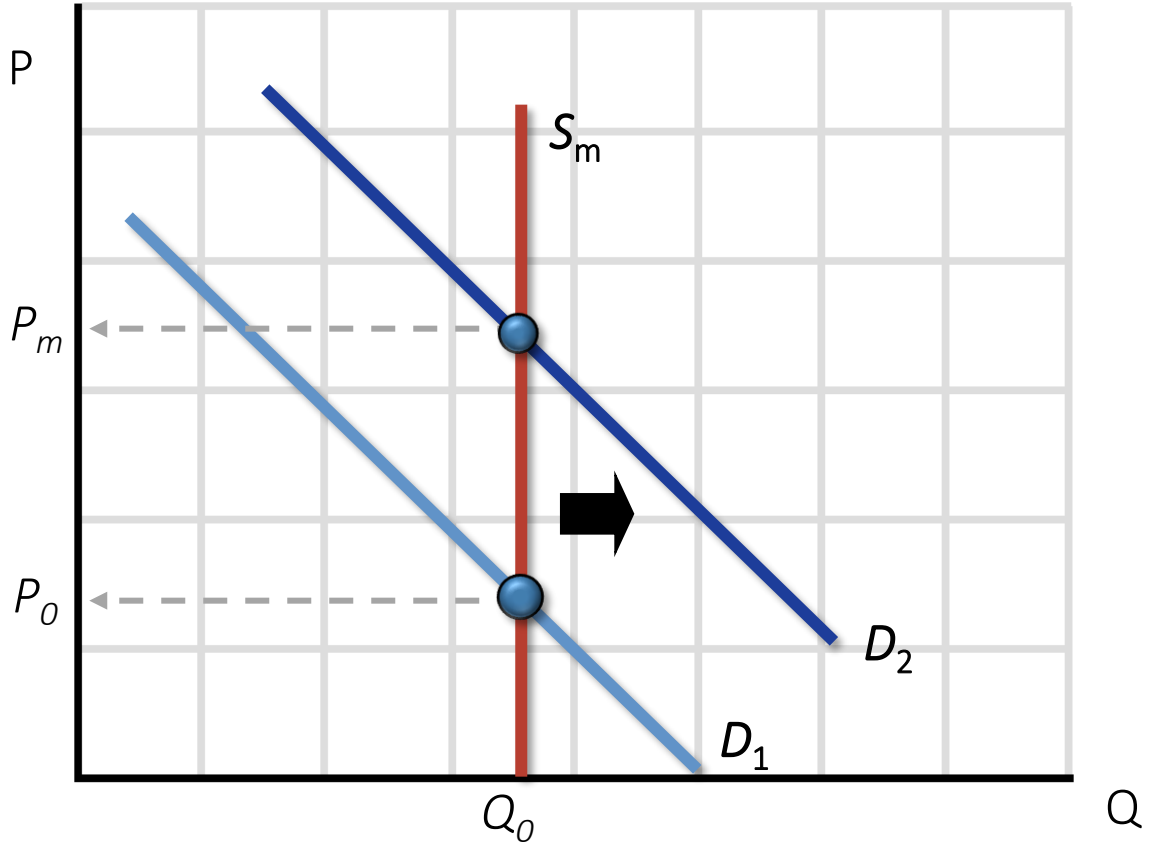
FIGURE 6-4 Time and Elasticity of Supply 2/6



Perfectly inelastic supply

(a) Immediate Market Period

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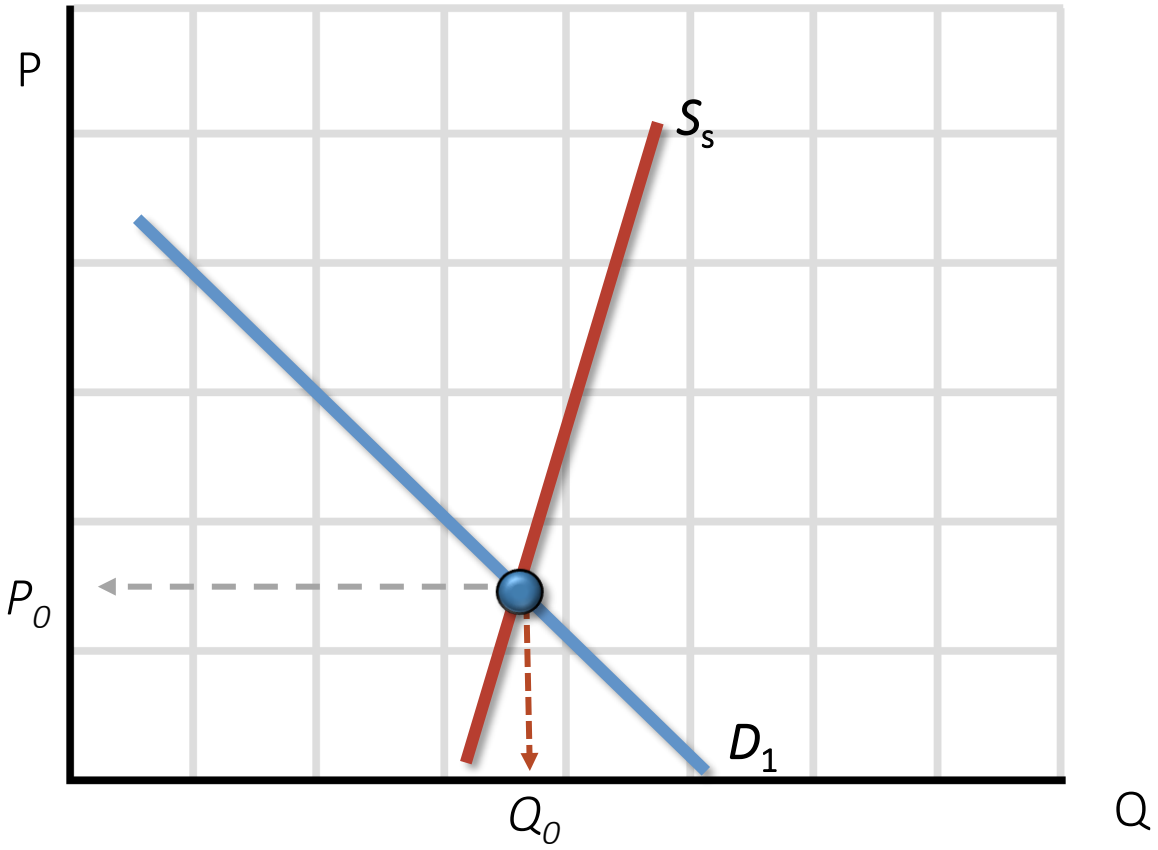
FIGURE 6-4 Time and Elasticity of Supply 3/6



Supply is more elastic than in market period

(b) Short-run

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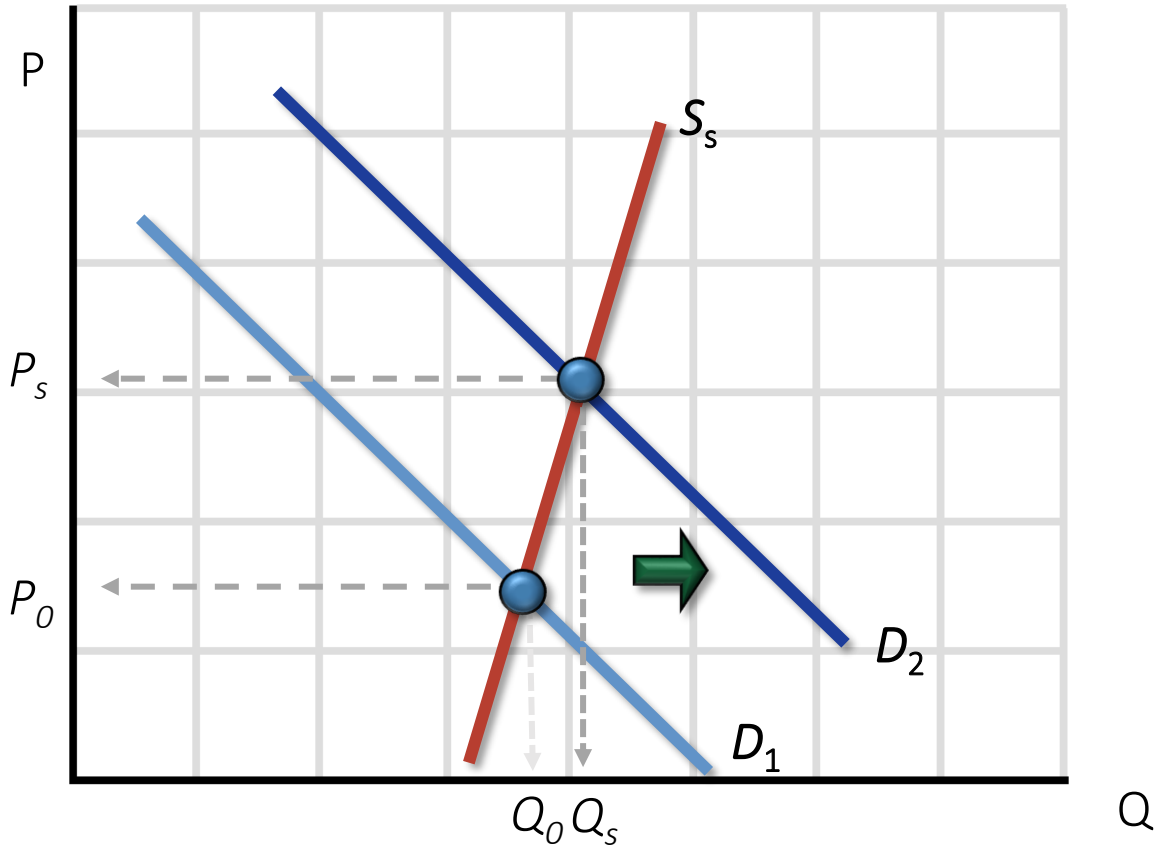
FIGURE 6-4 Time and Elasticity of Supply 4/6



Supply is more elastic than in market period

(b) Short-run

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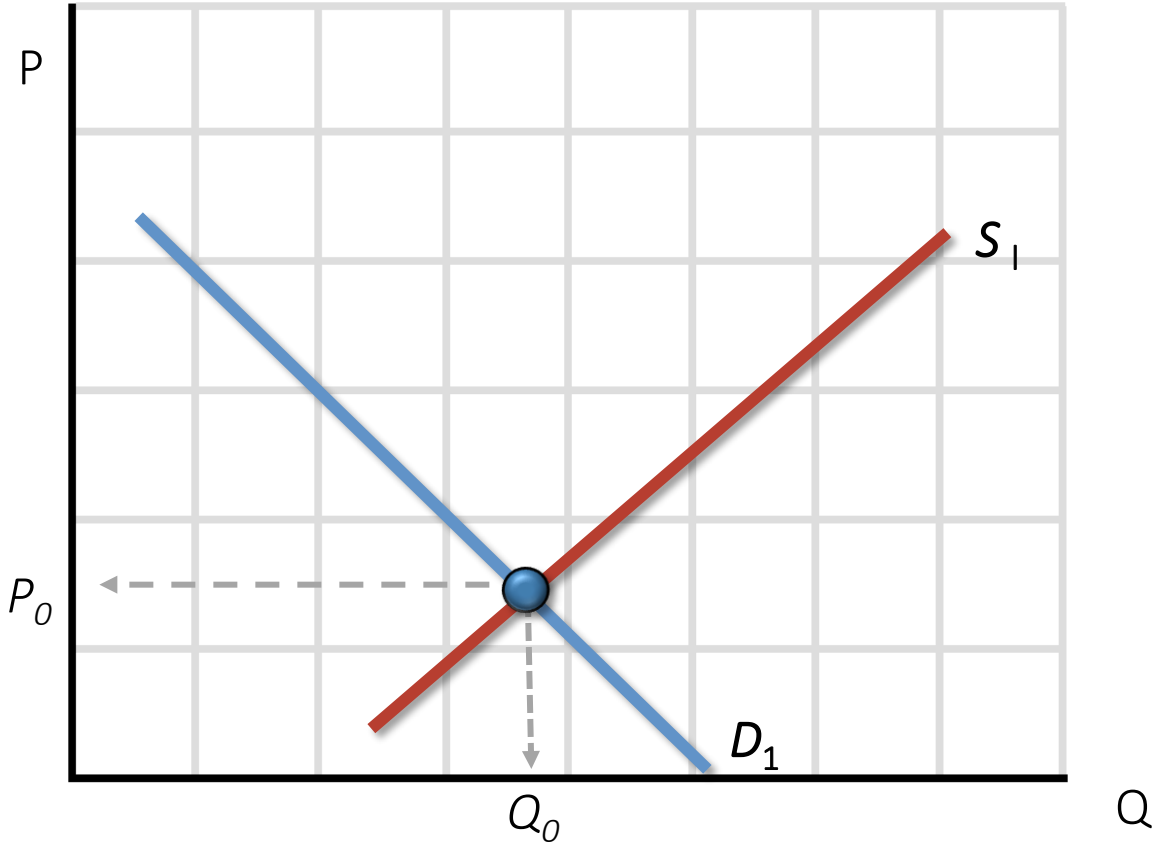
FIGURE 6-4 Time and Elasticity of Supply 5/6



Supply is even more elastic than in the long-run

(c) Long-run

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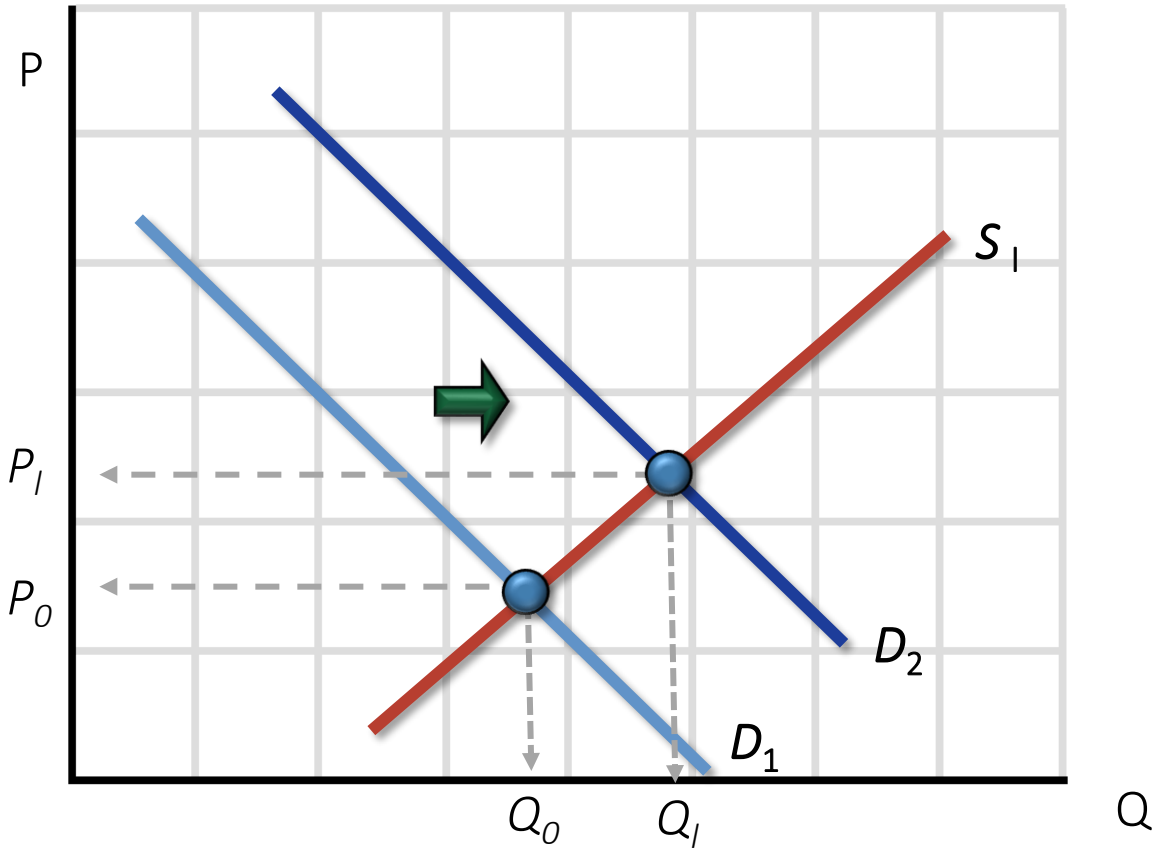
FIGURE 6-4 Time and Elasticity of Supply 6/6



Supply is even more elastic than in the long-run

(c) Long-run

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6.4 PRICE ELASTICITY OF SUPPLY 4/5

Applications of Price Elasticity

- Antiques:
  - The high prices of antiques result from strong demand and their highly inelastic supply since genuine antiques can't be reproduced.
  - Increased demand has greatly boosted their prices. In contrast, reproductions have an elastic supply, so their prices rise only slightly with increased demand.

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6.4 PRICE ELASTICITY OF SUPPLY 5/5

Applications of Price Elasticity

- Gold Prices:
  - Gold prices are volatile due to shifts in demand interacting with highly inelastic supply.
  - The costly and limited production of gold means that changes in demand result in large price fluctuations.

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6.5 CROSS ELASTICITY AND INCOME ELASTICITY OF DEMAND 1/6

Cross Elasticity of Demand

- Measures how responsive consumer purchases of one product (call it X) are to a change in the price of some other product (call it Y)

$$E_{xy} = \frac{\textit{Percentage change in quantity demanded of product X}}{\textit{Percentage change in the price of product Y}} = \frac{\%\Delta Q_x}{\%\Delta P_y}$$

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6.5 CROSS ELASTICITY AND INCOME ELASTICITY OF DEMAND 2/6

Cross Elasticity of Demand

Substitute Goods - positive sign

Complementary Goods - negative sign

Independent Goods - near zero

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6.5 CROSS ELASTICITY AND INCOME ELASTICITY OF DEMAND 3/6

Applications of Price Elasticity

- Cross elasticity helps businesses assess how a price change in one product affects the sales of another, like Sprite's effect on Coke sales.
- Low cross elasticity indicates weak substitutes.
- Governments use it to evaluate merger impacts on competition.
- Unrelated products with low cross elasticity rarely affect competition.

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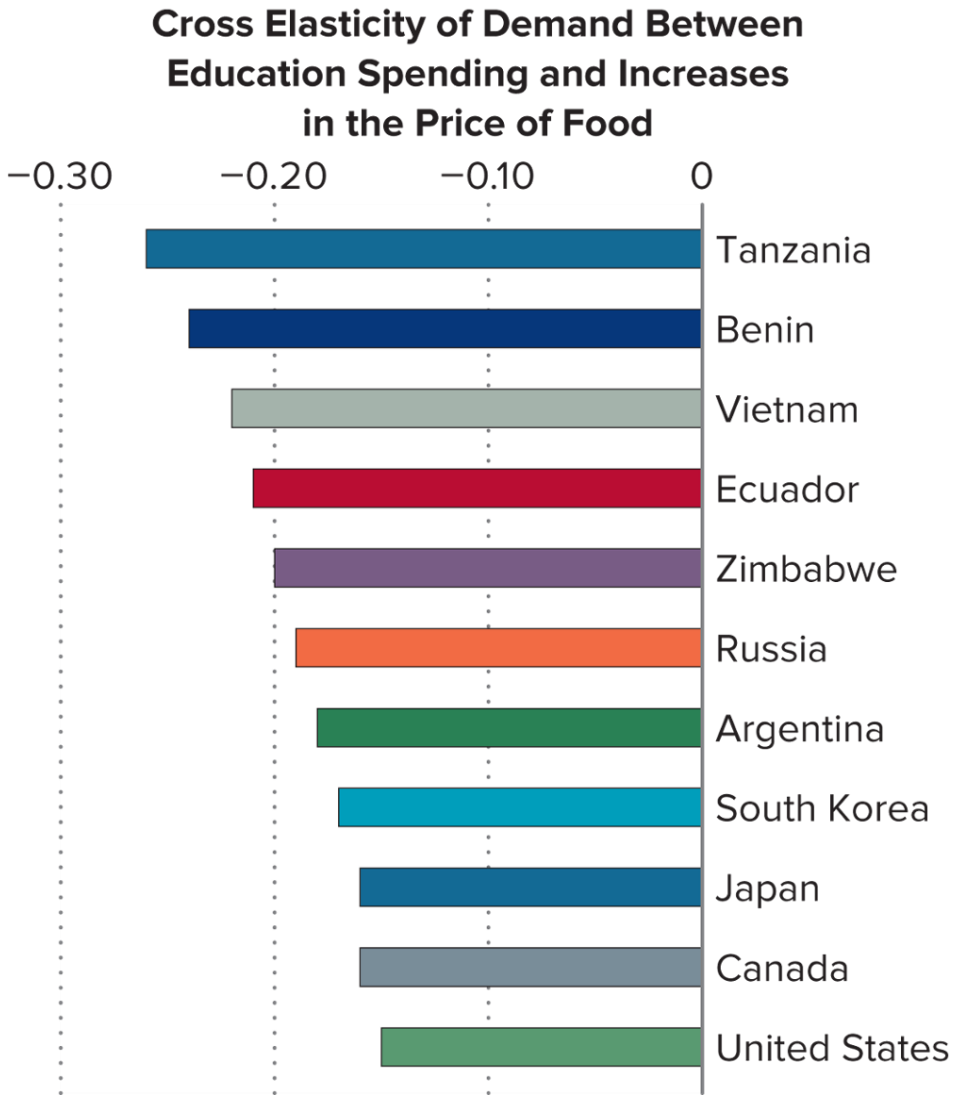
GLOBAL PERSPECTIVE 6.1

Cross Elasticity of Demand Between Education Spending and Increases in the Price of Food, Selected Nations

- Education spending drops more in low-income countries with rising food prices.
- Higher-income countries see a smaller impact on education spending.
- Tight family budgets amplify spending changes.



Source: Economic Research Service, United States Department of Agriculture.

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Income Elasticity of Demand

- Measures the degree to which consumers respond to a change in their incomes by buying more or less of a particular good.

$$E_i = \frac{\textit{Percentage change in quantity demanded of product X}}{\textit{Percentage change in income}} = \frac{\%\Delta Q_x}{\%\Delta I}$$

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6.5 CROSS ELASTICITY AND INCOME ELASTICITY OF DEMAND 5/6

- **Income Elasticity of Demand**

Normal Goods *- positive sign*

Inferior Goods *- negative sign*

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6.5 CROSS ELASTICITY AND INCOME ELASTICITY OF DEMAND 6/6

Income Elasticity of Demand: Insights

- High-income elasticity products, like cars and housing, decline rapidly in recessions.
- Low-income elasticity products, like home-prepared food, remain stable.
- Essentials have lower income elasticity than luxuries.
- Recessions impact luxury purchases more than essentials.

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Table 6-4 Cross Elasticity and Income Elasticity of Demand

Value of Coefficient	Description	Type of Good(s)
Cross elasticity:		
Positive ($E_{wz} > 0$)	Quantity demanded of W changes in same direction as change in price of Z	Substitutes
Negative ($E_{xy} < 0$)	Quantity demanded of X changes in opposite direction from change in price of Y	Complements
Income elasticity:		
Positive ($E_i > 0$)	Quantity demanded of the product changes in same direction as change in income	Normal or superior
Negative ($E_i < 0$)	Quantity demanded of the product changes in opposite direction from change in income	Inferior

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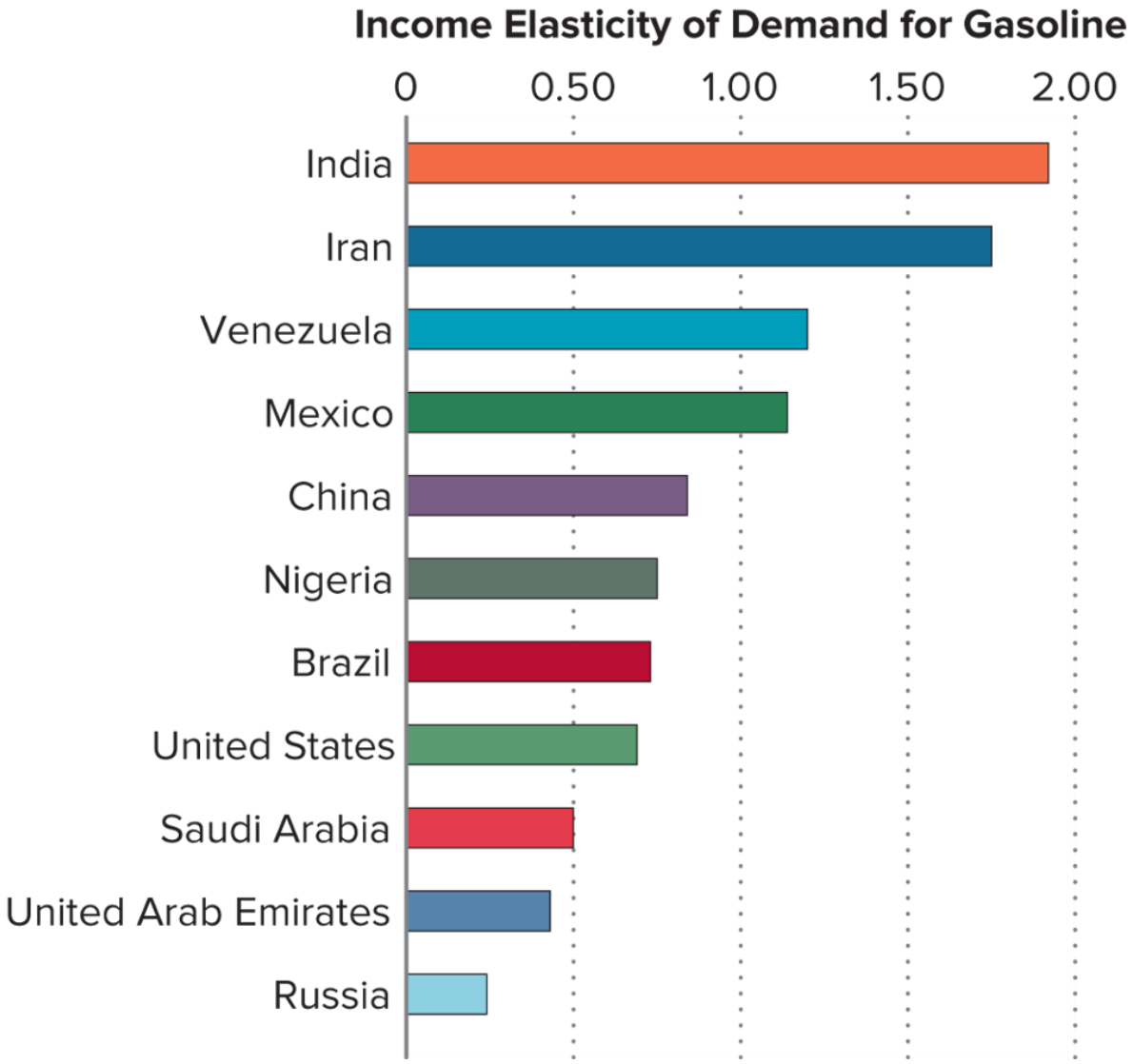
GLOBAL PERSPECTIVE 6.2

Income Elasticity of Demand for Gasoline, Selected Nations

- The income elasticity of gasoline demand varies widely across countries, with larger demand responses found in lower-income countries in which driving is a luxury for much of the population.



Source: U.S. Energy Information Administration.

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6.6 ELASTICITY AND TAX INCIDENCE 1/3

- Harmonized sales tax
  - Ontario, New Brunswick, Nova Scotia, and Newfoundland and Labrador
  - The HST varies across provinces
  - The HST extends provincial taxes on services
- Elasticity and Tax Incidence
  - Division of Burden

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FIGURE 6-5 The Incidence of a Tax 1/2

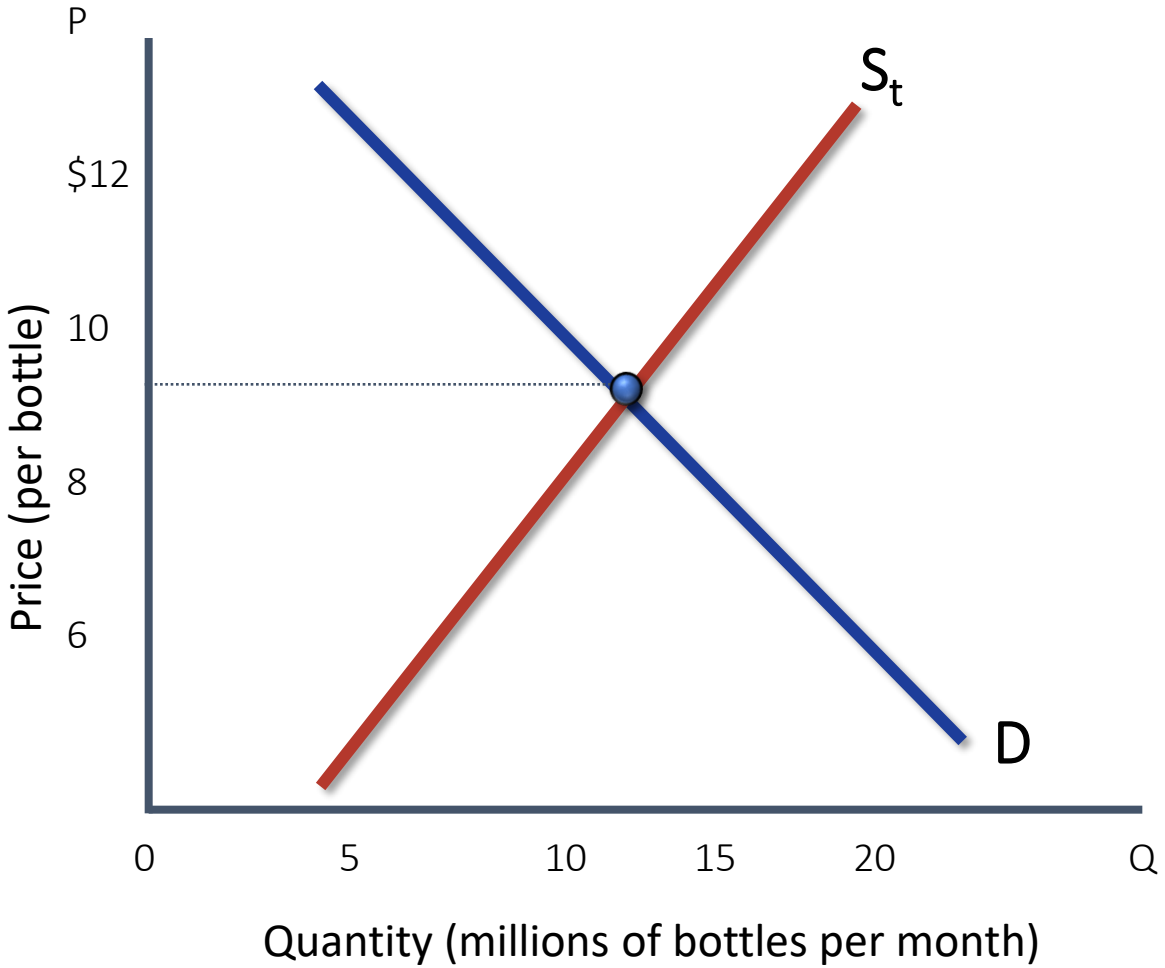

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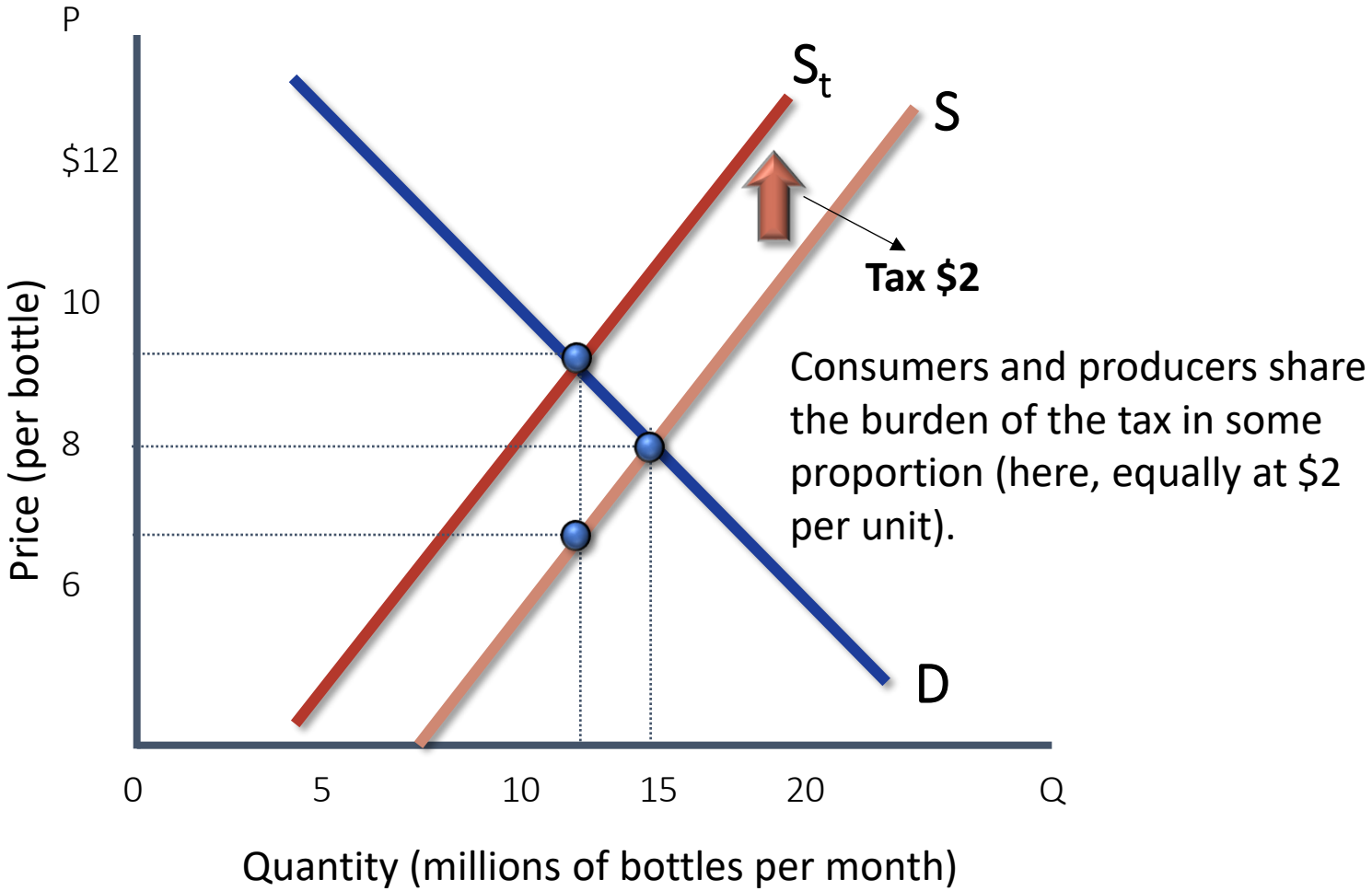
FIGURE 6-5 The Incidence of a Tax 2/2



Consumers and producers share the burden of the tax in some proportion (here, equally at $2 per unit).

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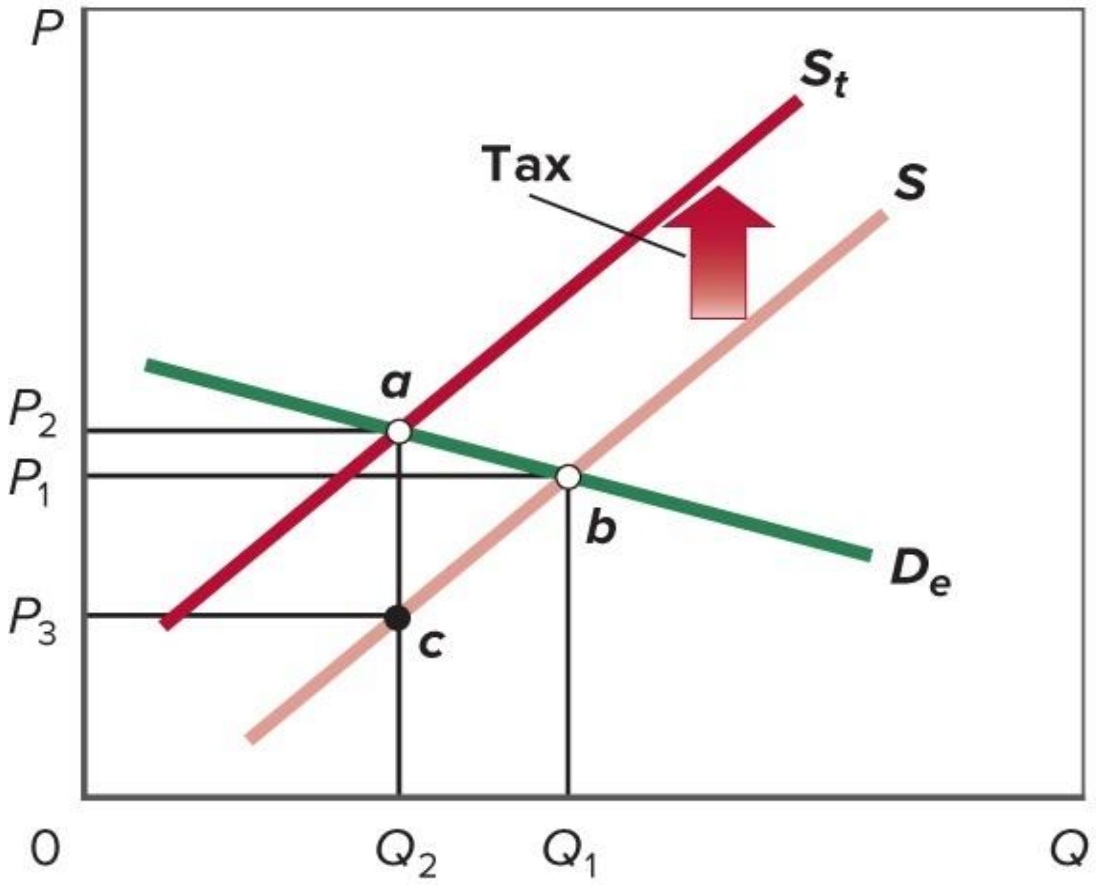
FIGURE 6-6 Demand Elasticity and the Incidence of a Tax 1/2

- If demand is elastic in the relevant price range, price rises modestly (P_1 to P_2) when a tax is levied.
- Hence, the producer bears most of the tax burden.

There is a typo in the book, rely on the slide



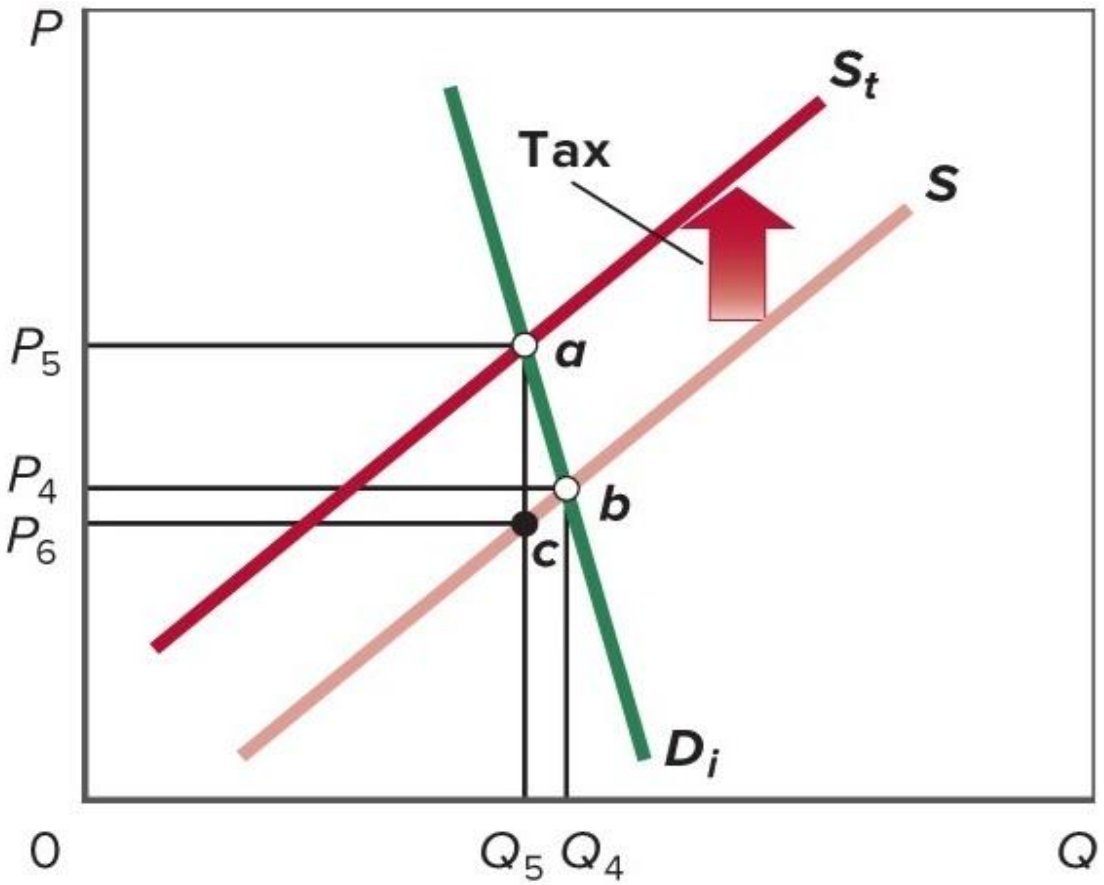
(a)

Tax incidence with elastic demand

FIGURE 6-6 Demand Elasticity and the Incidence of a Tax 2/2

- If demand is inelastic, the price to the buyer will increase substantially (P_4 to P_5).
- Most of the tax will be shifted to consumers.

There is a typo
in the book,
rely on the slide



(b)
Tax incidence with inelastic demand

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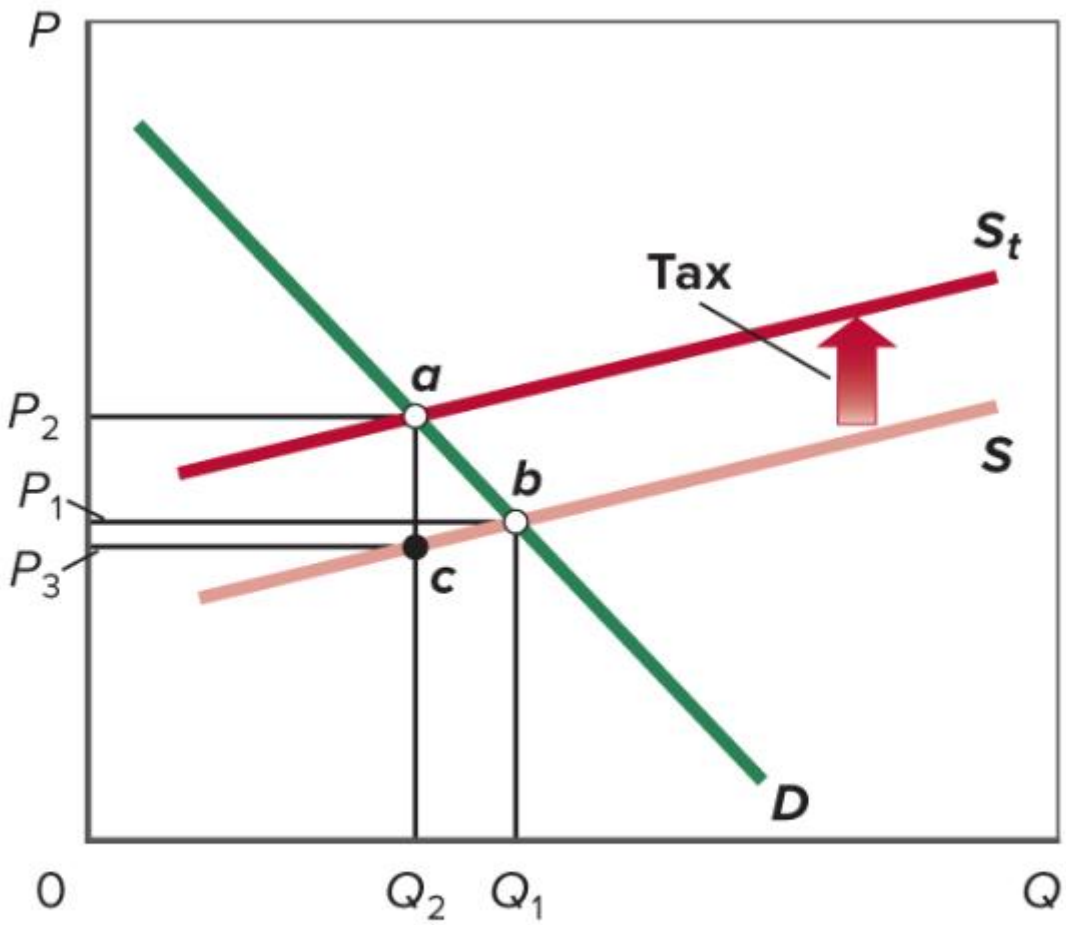
FIGURE 6-7 Supply Elasticity and the Incidence of a Tax 1/2

- With an elastic supply, the tax results in a large price increase (P_1 to P_2).
- The tax is therefore paid mainly by consumers.



(a)

Tax incidence with elastic supply

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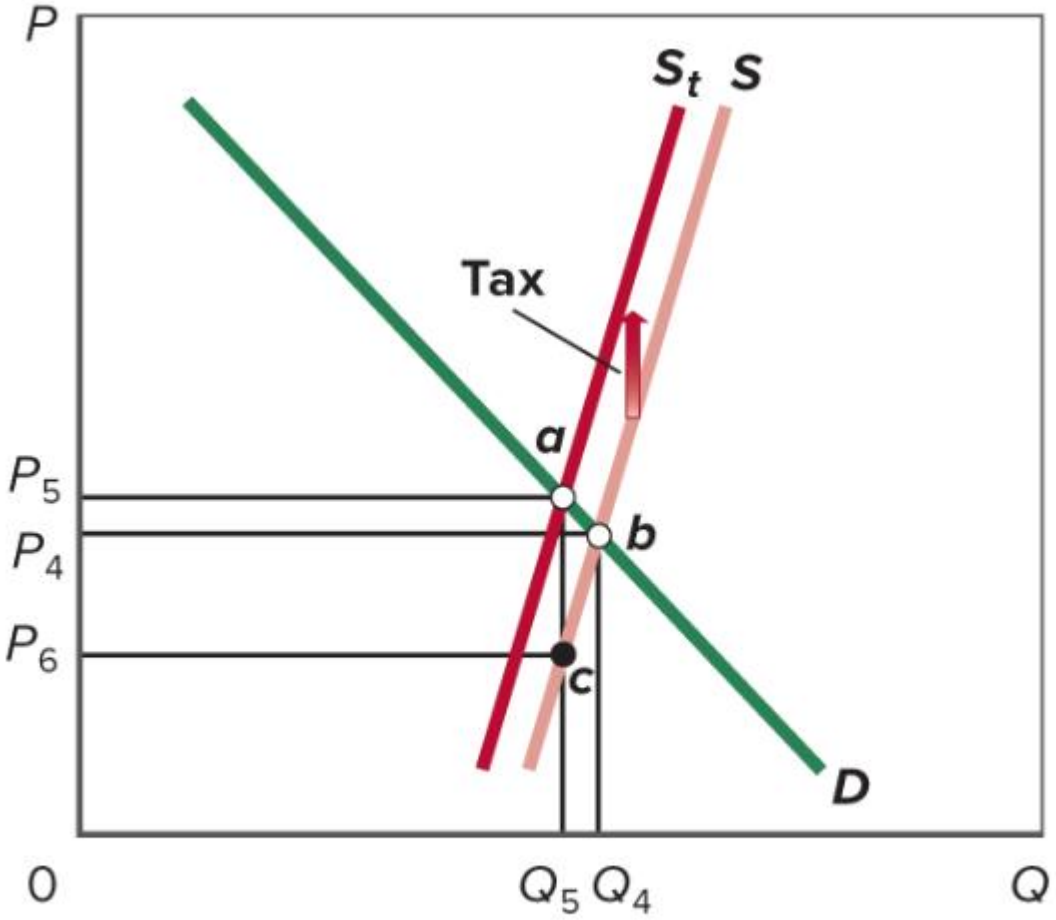
FIGURE 6-7 Supply Elasticity and the Incidence of a Tax 2/2

- If supply is inelastic, the price rise is small (P_4 to P_5).
- Sellers will have to bear most of the tax.



(b)

Tax incidence with inelastic supply

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6.6 ELASTICITY AND TAX INCIDENCE 2/3

Efficiency Loss of a Tax

- Tax Revenues
  - Efficiency Loss
  - Role of Elasticities
  - Qualifications
    - Redistributive goals
    - Reducing negative externalities

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6.6 ELASTICITY AND TAX INCIDENCE 3/3

Efficiency Loss of a Tax

- Greater supply/demand elasticities increase tax-induced efficiency loss.
- Taxes with equal revenue don't necessarily impose equal societal costs.
- Other tax goals include income redistribution and reducing negative externalities.
- Taxes on harmful goods like alcohol aim to reduce negative externalities (sin taxes).
- Sin taxes discourage harmful activities, such as alcohol and cigarette consumption.

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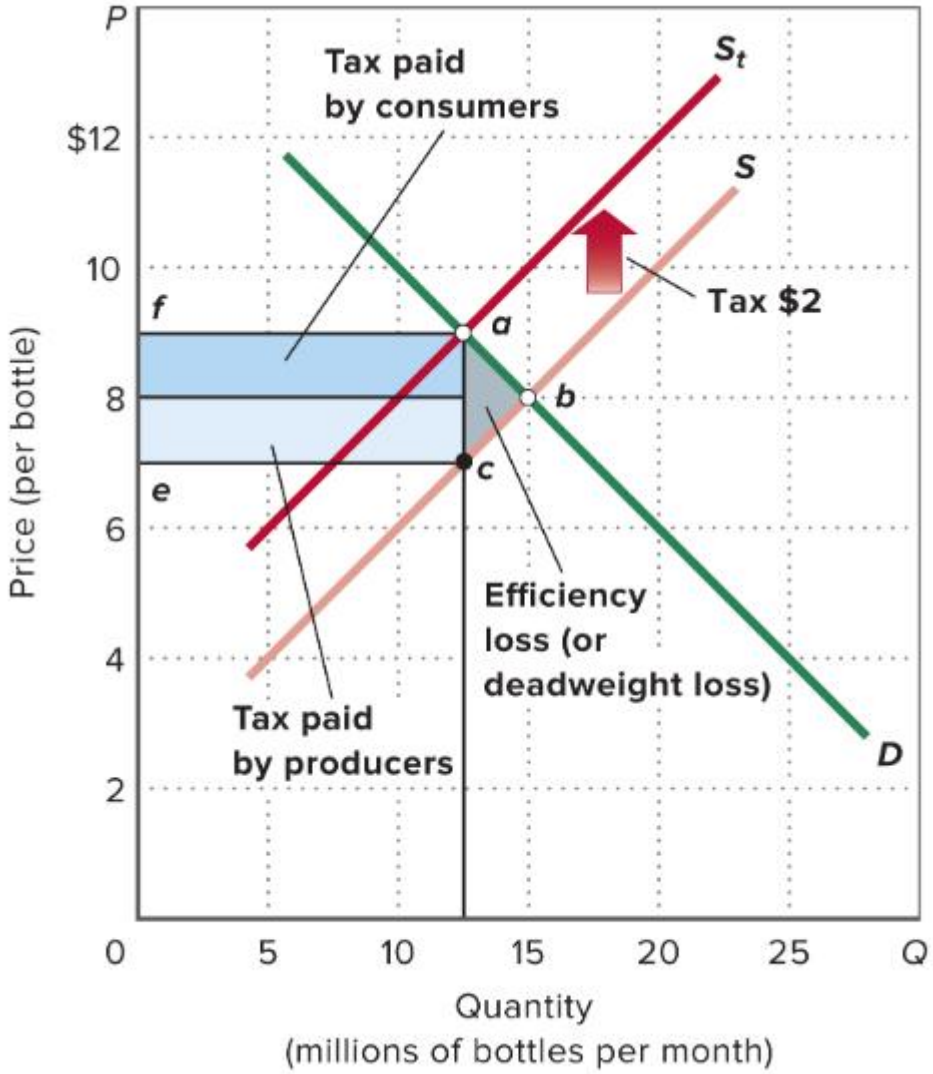
FIGURE 6-8 Efficiency Loss (or Deadweight Loss) of an Excise Tax Paid by Producers

- A $2 tax raises wine prices from $8 to $9, reducing quantity from 15M to 12.5M.
- Efficiency loss arises from the 2.5M output decline, a deadweight loss.
- Tax revenue ($25M) is split between producers and consumers based on elasticity.



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Elasticity and Pricing Power: Why Different Consumers Pay Different Prices

- Firms and nonprofit institutions often recognize and exploit differences in price elasticity of demand.

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THE LAST WORD 2/3

Elasticity and Pricing Power: Why Different Consumers Pay Different Prices

- Buyers pay the same market price regardless of individual demand elasticity.
- Firms with pricing power can charge different prices based on group elasticity.
- Business travellers face higher airfares due to their inelastic demand.
- Leisure travellers benefit from cheaper, restricted tickets due to their elastic demand.

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THE LAST WORD 3/3

Elasticity and Pricing Power: Why Different Consumers Pay Different Prices

- Children's discounts exist because they have more elastic demand than adults.
- Colleges use financial aid to charge lower-income students less due to higher elasticity.
- High-income students typically pay full tuition, supporting school revenue.
- Price discrimination strategies help firms maximize revenue based on demand elasticity.

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CHAPTER SUMMARY 1/4

- Price Elasticity of Demand: Measures consumer response to price changes; elastic if consumers are sensitive, inelastic if not.
- Coefficient Ed: Calculated as the percentage change in quantity demanded divided by the percentage change in price.
- Elasticity Values: Ed > 1 (elastic), Ed < 1 (inelastic), Ed = 1 (unit elasticity).

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CHAPTER SUMMARY 2/4

- Demand Curve: Elastic in the upper left, inelastic in the lower right; steepness does not indicate elasticity.
- Total-Revenue Test: If total revenue moves opposite to price, demand is elastic; the same direction means inelastic.
- Determinants of Price elasticity of demand: Number of substitutes, item's price relative to income, luxury vs. necessity, adjustment time.

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CHAPTER SUMMARY 3/4

- Elasticity of Supply: Calculated by percentage change in quantity supplied divided by percentage change in price; depends on resource mobility and time to adjust.
- Cross Elasticity: Measures the sensitivity of demand for one product to the price change of another; positive for substitutes, negative for complements.
- Income Elasticity: Shows demand responsiveness to income changes; positive for normal goods, negative for inferior goods.

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CHAPTER SUMMARY 4/4

- Industry Impact: High-income elasticity industries suffer more in recessions; low or negative elasticity industries fare better.
- Tax Incidence: More inelastic demand means consumers bear more tax; inelastic supply means sellers bear more.
- Efficiency Loss: Greater supply and demand elasticity increases the deadweight loss from taxation.

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